

Krishnamoorthi J

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Senior Research Fellow, Institute of Physics Bhubaneswar

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Research Interests

Neutrino oscillations, Earth tomography using atmospheric neutrinos, matter effects and non-standard interactions, detector physics, and data analysis in large-scale neutrino experiments (IceCube DeepCore, INO).

Education

PhD in Physics

Aligarh Muslim University, Aligarh

2023 – Present

Master of Science – Physics

Madurai Kamaraj University, Madurai

Dissertation: *Determination of Average Muon Lifetime and Energy Spectrum of Electron from Muon Decay*

2017 – 2019

Bachelor of Science – Physics

Thiagarajar College, Madurai

2014 – 2017

Research Experience

Senior Research Fellow

Institute of Physics, Bhubaneswar

2024 – Present

Junior Research Fellow

Institute of Physics, Bhubaneswar

2022 – 2024

Project Scientific Officer (B) [Detector]

Tata Institute of Fundamental Research, Mumbai

2021 – 2022

Publications

- **IceCube Collaboration**, *Estimating the sensitivity of the IceCube Upgrade to probe the interior of the Earth using atmospheric neutrino oscillations*, arXiv:2608.06543 [hep-ex].
- Gopal Garg, **Krishnamoorthi J.**, Anil Kumar, Sanjib Kumar Agarwalla, *Constraints on long-range neutrino interactions from a variety of $U(1)$ ' symmetries using atmospheric neutrinos at IceCube DeepCore*, arXiv:2606.03443 [hep-ph].
- **Krishnamoorthi J.**, Anil Kumar, Sanjib Kumar Agarwalla, *δ_{CP} -free constraints on NSI parameters $\varepsilon_{e\mu}$ and $\varepsilon_{e\tau}$ using high-purity ν_μ CC events at IceCube DeepCore*, Physical Review D 114, 015007 (2026), arXiv:2604.16157 [hep-ph].
- Gopal Garg, **Krishnamoorthi J.**, Anil Kumar, Sanjib Kumar Agarwalla, *First Constraints on Long-Range Neutrino Interactions using IceCube DeepCore*, arXiv:2601.01220 [hep-ph].

- **Krishnamoorthi J.**, Anil Kumar, Sanjib Kumar Agarwalla, *Stringent constraints on non-standard neutrino interactions using high-purity ν_μ CC events in IceCube DeepCore*, arXiv:2512.22632 [hep-ph].
- **Krishnamoorthi J.**, Anuj Kumar Upadhyay, Anil Kumar, Sanjib Kumar Agarwalla, *Probing Earth's core using atmospheric neutrino oscillations in the presence of NSI at INO-ICAL*, Journal of High Energy Physics 213 (2025).
- Mandar Saraf et al., *Design, fabrication and large-scale qualification of cosmic muon veto scintillator detectors*, Journal of Instrumentation 18, P05003 (2023), arXiv:2301.12486 [physics.ins-det].

Conference Proceedings

- **Krishnamoorthi J.**, *Exploring Layered Structure Inside Earth Using Atmospheric Neutrino Oscillation at IceCube DeepCore*, Springer Proc. Phys. 322 (2026) 447–450.
- Sharmistha Chattopadhyay, **Krishnamoorthi J.**, Anuj Kumar Upadhyay, *Demonstrating the ability of IceCube DeepCore to probe Earth's interior with atmospheric neutrino oscillations*, European Physical Journal Special Topics 234, 5055–5064 (2025).

Talks, Posters, and Schools

- Poster: *Neutrino Oscillations as a Window to the Earth's Interior*, IIFC- ν P School, NISER Bhubaneswar, October 2025.
- Poster: *Neutrino Tomography: A Window into Earth's Interior*, ASAR 2025, KIIT Bhubaneswar, March 2025.
- Contributed Talk: *Validating Layered Structure Inside Earth Using Neutrino Oscillations at IceCube DeepCore*, XVII International Conference on Interconnections between Particle Physics and Cosmology, IIT Hyderabad, October 2024.
- Poster: *Exploring the Interior of Earth Using Atmospheric Neutrino Oscillations at IceCube DeepCore*, 1st DAE Conclave, NISER, October 2024.
- Poster: *Exploring Earth's Interior with Atmospheric Neutrinos at IceCube DeepCore*, Radio Neutrino & Cosmic Rays Astronomy Workshop, AMU, April 2024.

Internship

Visiting Students' Research Program (VSRP)

2018

Department of High Energy Physics, Tata Institute of Fundamental Research, Mumbai

Technical Skills

Programming	C++, Python, Bash
Scientific Computing	NumPy, SciPy, Pandas
HEP / Neutrino Analysis	ROOT, GENIE, PISA, LeptonInjector
Computing	Linux, Git, HPC, HTCondor, Slurm
Experimental	Particle detectors, detector characterization

Academic Service & Outreach

- Instructor, ICFA Instrumentation School, TIFR Mumbai (February 2023).

- Organizer and Resource Person, IceCube Masterclass, Institute of Physics Bhubaneswar (March 2025).
- Designed and built a functional 3D LED model of the IceCube detector capable of visualizing neutrino interaction events and light propagation patterns.